## Digestion in mouth

The digestion of fat starts in the mouth by the action of Lingual lipase, which has an optimum pH between 4.5 and 5.4. It acts on short chain triglycerides that are typically present in milk, butter and ghee. The action of lingual lipase is more important for new born infants and children (Table 2.2).

**Table 2.2 Various lipases and the site of action**

| Lipase | Site of action | Preferred substrate | Products (s) |
|---|---|---|---|
| Lingual/acid - stable lipase | Mouth, stomach | TAGs with medium / short chain FAs | FFA+DAG |
| Pancreatic lipase + co-lipase | Small intestine | TAGs with long - chain FAs | 2 FFA+MAG |
| Intestinal lipase with bile acids | Small intestine | TAGs with medium chain FAs | 3 FFA+ glycerol |
| Phospho-lipase $A_2$ + bile acids | Small intestine | PLs with unsat. FA on position 2 | Unsat FFA + lysolecithin |

## Digestion in stomach

Gastric lipase with an optimum pH of 4 -5 is secreted into the stomach by the chief cells. This also acts on breaking down fat. However, the gastric lipase digests the fat to a minimal extent because of the prevailing high acidic pH and absence of emulsification. But, unlike pepsin, it is not dependent on an acid pH, and therefore remains active in the small intestine, and constitutes about 30% of the total lipase secreted over a 3-hour period. Gastric lipase is not essential in fat digestion, but resulting fatty acids and peptides formed by pepsin help to coordinate gastric emptying and pancreatic secretion.

**Satiety:** Fats have a very high satiety value ( fullness of the stomach), by inhibiting the gastric motility ( movement of the chyme from stomach to duodenum). This effect of dietary lipids is mediated by the hormone enterogastrone.

## Digestion in duodenum

The bile (pH 7.7) entering the duodenum serves to neutralize the acid chyme from the stomach and provides a pH favorable for the action of pancreatic enzymes and small intestine (Figure 2.19).

The hydrophobic regions of bile salts intercalate into the lipid aggregates, such that the hydrophilic domains face the surface resulting in the breakdown of larger aggregates into small droplets. This increases the surface area for action of lipase.

The lipolytic enzymes present in the pancreatic secretions are:



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